

Introduction

CPT204: Advanced Object-Oriented Programming

Lecture 1.0

Welcome to CPT204!

Advanced Object-Oriented Programming

This course will:

- Introduce **data structures** and **advanced programming language features** within the context of Java



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Coursework and Exam

- CW1: Two In-class Quizzes (20%)
 - **Two 50-minute open-book tests** with invigilation, in Week 5 & 9's labs
- Group Project (40%)
 - **Two students** work together to **solve the problem** and present their work in **report** and **video recording**
 - Groups will be assigned around **Week 8**
 - The task sheet is expected to be released in **Week 9** with the deadline being **one month later**
- Final Exam (40%)
 - 2-hours, open-book on LMO, on Safe Exam Browser (SEB)
 - Permitted to bring lecture/tutorial notes and .java files on USB to the exam

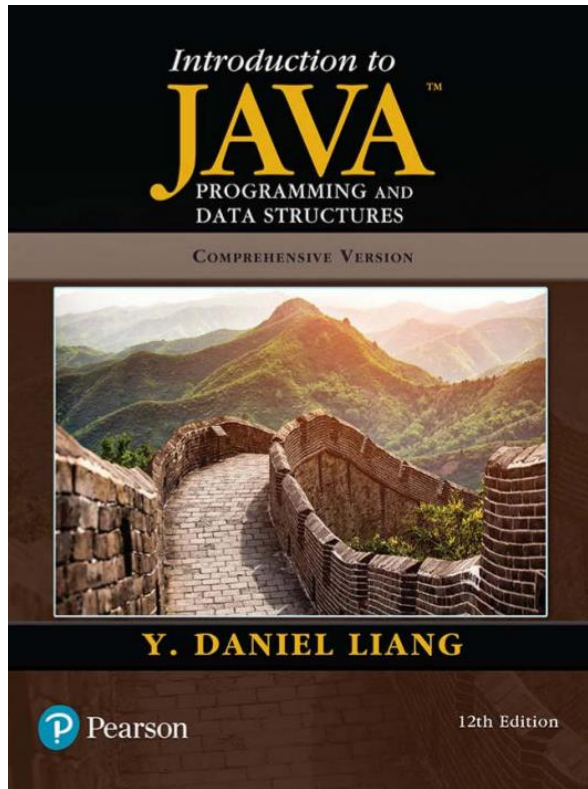
Coursework and Exam

- Resit (100%)
 - Work on a **team-based large programming project**
 - **Two students** work together to **solve the problem** and present their work in **report** and **video recording**
 - Timeline details will be announced in the summer

Weekly Time Commitment

- **Two hours of lecture** per week
- **Two hours of supervised lab sessions** once per week
 - **Practice your skills** and **seek help from TAs**
- **Self-study** time (~8 hours per week)
 - Private study (reading, coding practice, preparation for in-class quizzes and final exam)
 - Work on group project

Textbook



Liang Y. Daniel - **Introduction to Java Programming and Data Structures** -12th edition

- You can find this [on Learning Mall](#)

Teaching Team



Yihong Wang
(Module Leader)

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Office Hours: Mondays
15:00-16:00, Tuesdays 13:00-
14:00, SD423



Bo Zhao

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Office Hours: Wednesdays
14:00-16:00, Thursdays 14:00-
16:00, SC544A



Brennan Jones

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Office Hours: Mondays
15:00-16:00, Thursdays 14:00-
15:00, SC564D

Topics

Week	Topic	Teacher	Pre-reading and others
W1	Review Java: Arrays, OOP 1	Brennan Jones	Liang Ch. 7, 9, 10
W2	Review Java: OOP 2	Brennan Jones	Liang Ch. 10, 11
W3	Abstract Classes and Interfaces	Brennan Jones	Liang Ch. 13
W4	Generics	Brennan Jones	Liang Ch. 19
W5	Lists, Stacks, Queues, and Priority Queues	Bo Zhao	Liang Ch. 20
W6	Sets and Maps	Bo Zhao	Liang Ch. 21
W7	<i>Non-teaching week</i>	--	--
W8	Developing Efficient Algorithms	Bo Zhao	Liang Ch. 22
W9	Sorting	Yihong Wang	Liang Ch. 23
W10	Graphs, Weighted Graphs	Yihong Wang	Liang Ch. 28, 29
W11	Binary Search Trees	Yihong Wang	Liang Ch. 25
W12	AVL Trees and Hashing	Yihong Wang	Liang Ch. 26, 27
W13	Revision	Yihong Wang, Bo Zhao, Brennan Jones	All topics